

IZUM ADNAN

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EDUCATION

Columbia University

Master's of Science in Computer Science (MSCS) – Specialization in Machine Learning

New York City, NY

Expected Dec 2027

- Courses: Databases, Computation & the Brain (ML Models), AI-Accelerated Startup Lab, AI Cancer Research

Western University

Bachelor of Science – Honors Specialization in Computer Science, GPA: 3.81/4.0 (Dean's Honors List)

London, Ontario

Sep 2021 - May 2026

- Courses: AI, Analysis of Algorithms, Data Structures & Algorithms, Info Systems and Programming, Software Engineering

WORK EXPERIENCE

Celestica

Toronto, Ontario

Robotic Process Automation Developer Co-op

May 2024 - Aug 2025

- Led two enterprise-grade RPA deployments integrating Blue Prism, SAP, and Azure Bots for global manufacturing operations
- Built a business-analytics automation processing high-volume SAP inputs with data-integrity validation and cross-department conditional workflows, raising projected annual savings from ~\$80K to ~\$95K
- Developed an endpoint-validation automation that continuously tests live production endpoints, cutting RPA downtime incidents by 70%
- Partnered with cross-functional teams across North America, Europe, and Asia to map workflows and convert inefficiencies into scalable automation

Western AI

London, Ontario

AI Developer & Architect

Sep 2022 - Mar 2023

- Collaborated with a multidisciplinary team on applied machine learning, agent behavior, and game intelligence research
- Led end-to-end development of a competitive AI agent as project lead for a team of 8, driving architecture, testing, and execution
- Presented team's work at a 3-day multi-university conference, placing 2nd; architecture was later reused internally as a template for future agent-based research

PROJECTS

Adaptive Educational Game for Learners with Special Needs – Wellbotics

Sep 2026 – Present

- Building an interactive cardinality game for middle school students with special needs, as part of a game design cohort in partnership with Teachers College, Columbia
- Testing tangible gameplay with students and educators to evaluate usability and accessibility, adjusting the design based on their interactions and feedback

AI Driven Scoping Review Pipeline for Physiotherapy Research

Sep 2025 – May 2026

- Designed and integrated an end-to-end scoping review automation pipeline in Python (PubMed API, Pandas, Scikit-learn, spaCy), replacing a manual multi-stage workflow
- Trained an SVM + TF-IDF classifier for title-and-abstract screening, reaching 0.91 recall and 0.82 F1 across 1,247 records
- Automated structured data extraction with rule-based NER, populating key fields for 78% of included articles without manual input
- Cut total reviewer time by 64% (38 to 13.5 hours) versus a two-reviewer baseline while achieving reproducible outputs

Whac-A-Mole Game – Hardware Project

Sep 2023 – Nov 2023

- Produced an embedded interactive Whac-A-Mole game merging LED matrices, GPIO input, and real-time timing logic, using state machines for system flow and scoring algorithms with variable difficulty scaling
- Conducted performance testing to optimize response windows and measure average user interaction speed

Western Geographic Information System – Desktop Application

Sep 2022 – Dec 2022

- Programmed a Java/JavaFX indoor building navigation app using Western's GIS datasets, leveraging real-time OpenWeather API data for environmental context and implementing layered spatial data structures for pathfinding across buildings and indoor areas

SKILLS & CERTIFICATIONS

Technical Skills: Python, C, C++, PHP, Java, JavaScript, HTML, CSS, R, MySQL, MS Access, Pandas, scikit-learn, spaCy, data modeling, Microsoft Azure, Blue Prism (RPA), Microsoft Office Suite

Soft Skills: Cross-functional collaboration, stakeholder relationship management, communication, end-to-end project execution

Certifications: IBM Z XPLORE, AWS Certified Cloud Practitioner, Blue Prism Foundations (BP3 Global), IBM Software Engineering (Back-End), Duke Java & Software Engineering Specialization, Harvard CS50 Game Development